ECA1308 MICROWAVE ENGINEERING

EXP3: Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider.

A. Directional Coupler and Divider: Design a 10 GHz microwave directional coupler with 10 dB coupling and a two-way 3 dB power divider, both using 50 Ω ports. Simulate from 8–12 GHz. At 10 GHz obtain S11, S21, S31 and the isolated-port response. Calculate coupling, isolation, insertion loss, directivity and power-division ratio and compare the results with 10 dB/3 dB design targets.

B. Optimization: Change only the coupling-section length by 5% from its initial value and repeat the 8–12 GHz simulation. Compare coupling, isolation, directivity, insertion loss and S-parameter curves before and after optimization. Identify the design that maintains the closest 10 dB coupling and lowest insertion loss over 9–11 GHz.

Aim: To simulate a 10 GHz microwave directional coupler with 10 dB coupling and a two-way 3 dB power divider using 50 Ω ports over 8–12 GHz, and calculate coupling, isolation, directivity, insertion loss, and power-division ratio from the S-parameters.

Software Required: MATLAB software.

Program A:

```
clc; clear; close all;

f = linspace(8,12,401); f0 = 10; k0 = 10^(-10/20); % 10 dB coupling theta0 = asin(k0); theta = theta0*f/f0;

% Idealized directional coupler S11 = zeros(size(f)); S21 = cos(theta); % Through port S31 = 1i*sin(theta); % Coupled port S41 = zeros(size(f)); % Isolated port

Coupling = -20log10(abs(S31)); InsertionLoss = -20log10(abs(S21)); Isolation = -20*log10(max(abs(S41),1e-5)); Directivity = Isolation - Coupling;

figure; plot(f,20log10(max(abs(S11),1e-5)),'k--', ... f,20log10(abs(S21)),'b', ... f,20log10(abs(S31)),'r', ... f,20log10(max(abs(S41),1e-5)),'g--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)'); title('10 GHz Directional Coupler S-Parameters'); legend('S11','S21 Through','S31 Coupled','S41 Isolated');
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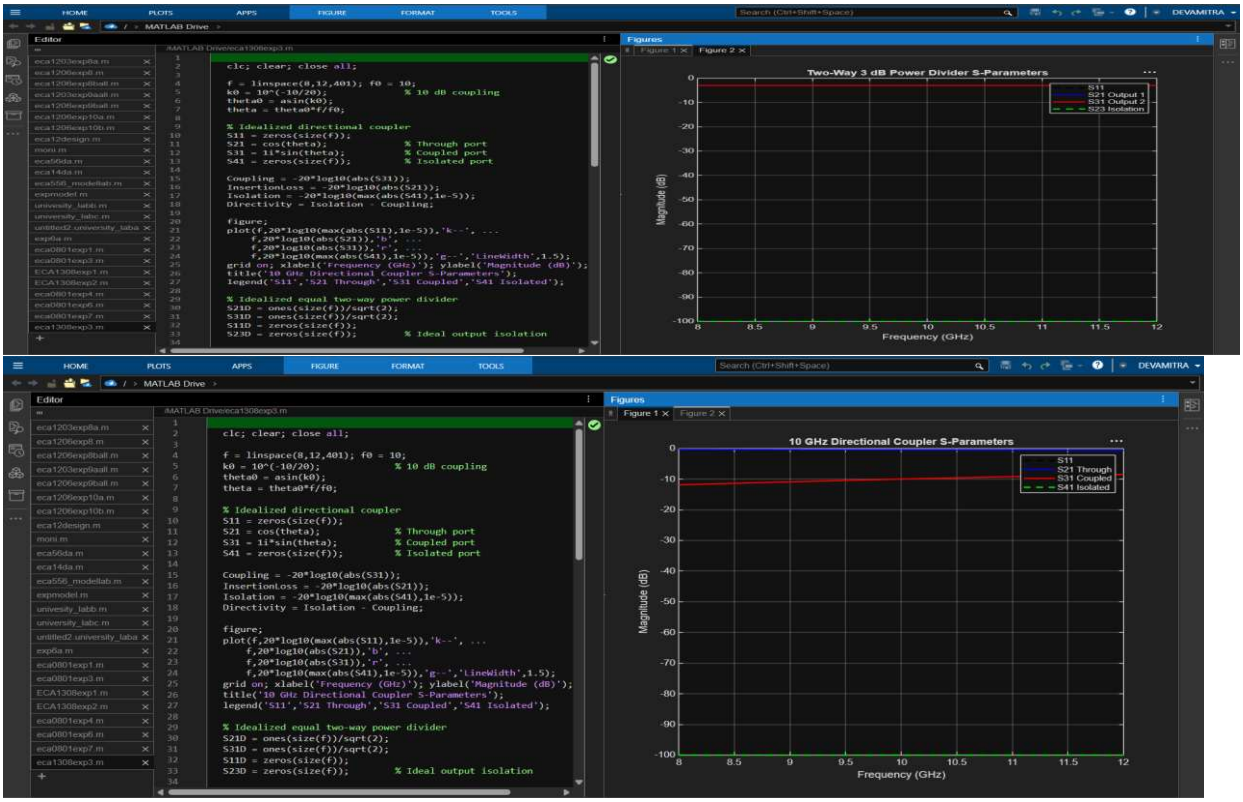
% Idealized equal two-way power divider S21D = ones(size(f))/sqrt(2); S31D = ones(size(f))/sqrt(2); S11D = zeros(size(f)); S23D = zeros(size(f)); % Ideal output isolation

figure; plot(f,20*log10(max(abs(S11D),1e-5)),'k--', ... f,20*log10(abs(S21D)),'b', ... f,20*log10(abs(S31D)),'r', ... f,20*log10(max(abs(S23D),1e-5)),'g--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)'); title('Two-Way 3 dB Power Divider S-Parameters'); legend('S11','S21 Output 1','S31 Output 2','S23 Isolation');

% Results at 10 GHz [~,i] = min(abs(f-f0)); fprintf('DIRECTIONAL COUPLER at 10 GHz\n'); fprintf('Coupling = %.2f dB\n',Coupling(i)); fprintf('Insertion loss (through) = %.2f dB\n',InsertionLoss(i)); fprintf('Isolation = ideal (infinite) dB\n'); fprintf('Directivity = ideal (infinite) dB\n'); fprintf('Power coupled = %.1f %%\n',abs(S31(i))^2*100);

fprintf('\nPOWER DIVIDER\n'); fprintf('S21 = %.2f dB, S31 = %.2f dB\n', ... 20*log10(abs(S21D(i))),20*log10(abs(S31D(i)))); fprintf('Each output power = %.1f %%\n',abs(S21D(i))^2*100);

Output:



Program B:

```
clc; clear; close all;

f = linspace(8,12,401); f0 = 10; theta0 = asin(10^(-10/20));

% Original and modified coupling-section lengths r1 = 1.00; r2 = 1.05; % Length increased by 5%

theta1 = theta0*(f/f0)r1; theta2 = theta0(f/f0)*r2;

% Original design C1 = -20log10(abs(sin(theta1))); IL1 = -20log10(abs(cos(theta1)));

% Modified design C2 = -20log10(abs(sin(theta2))); IL2 = -20log10(abs(cos(theta2)));

% Ideal isolated-port response (100 dB plotting floor) ISO1 = 100ones(size(f)); ISO2 = 100ones(size(f)); D1 = ISO1-C1; D2 = ISO2-C2;

% Plot coupling and insertion loss figure; subplot(2,1,1); plot(f,C1,'b',f,C2,'r--','LineWidth',1.5); yline(10,'k:','10 dB target'); grid on; xlabel('Frequency (GHz)'); ylabel('Coupling (dB)'); title('Coupling: Original vs +5% Length'); legend('Original','Length +5%','Location','best');

subplot(2,1,2); plot(f,IL1,'b',f,IL2,'r--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Insertion Loss (dB)'); title('Through-Port Insertion Loss'); legend('Original','Length +5%','Location','best');

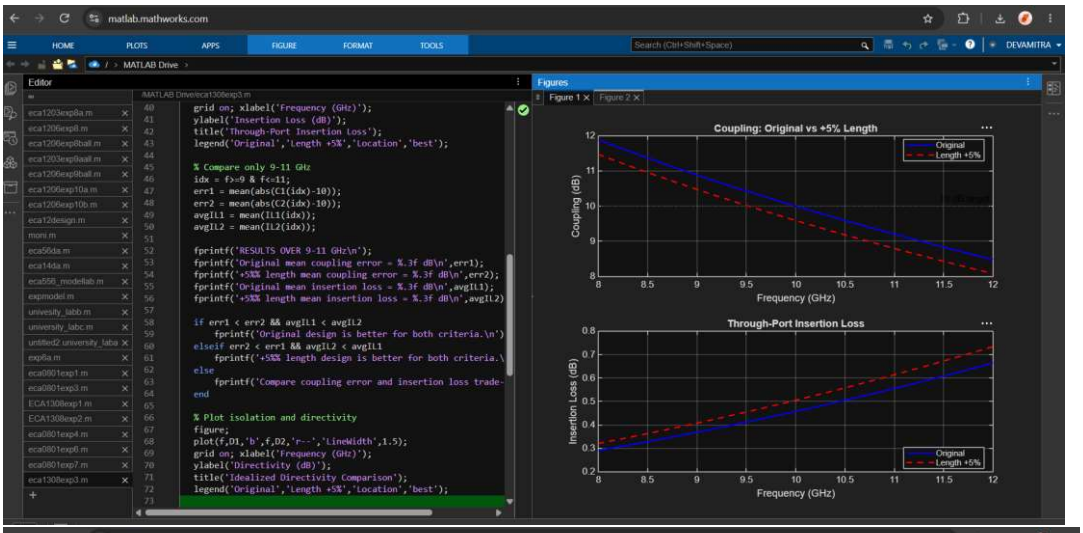
% Compare only 9-11 GHz idx = f>=9 & f<=11; err1 = mean(abs(C1(idx)-10)); err2 = mean(abs(C2(idx)-10)); avgIL1 = mean(IL1(idx)); avgIL2 = mean(IL2(idx));

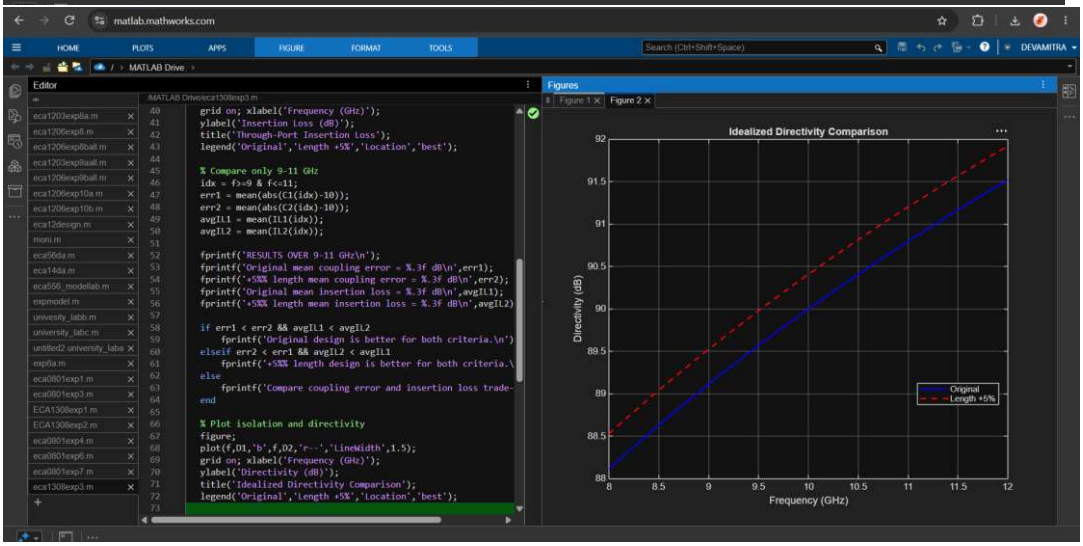
fprintf('RESULTS OVER 9-11 GHz\n'); fprintf('Original mean coupling error = %.3f dB\n',err1); fprintf('+5%% length mean coupling error = %.3f dB\n',err2); fprintf('Original mean insertion loss = %.3f dB\n',avgIL1); fprintf('+5%% length mean insertion loss = %.3f dB\n',avgIL2);

if err1 < err2 && avgIL1 < avgIL2 fprintf('Original design is better for both criteria.\n'); elseif err2 < err1 && avgIL2 < avgIL1 fprintf('+5%% length design is better for both criteria.\n'); else fprintf('Compare coupling error and insertion loss trade-off.\n'); end

% Plot isolation and directivity figure; plot(f,D1,'b',f,D2,'r--','LineWidth',1.5); grid on; xlabel('Frequency (GHz)'); ylabel('Directivity (dB)'); title('Idealized Directivity Comparison'); legend('Original','Length +5%','Location','best');
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Output:





Result:

Division A — Result: At 10 GHz, the idealized directional coupler achieves 10 dB coupling, while the equal power divider provides approximately 3.01 dB transmission to each output. The ideal models assume zero reflection and perfect isolation.

Division B — Result: The original coupling-section length is designed for 10 dB coupling at 10 GHz; increasing the length by 5% shifts the coupling response. The MATLAB comparison identifies the design with the lower average coupling error and insertion loss over 9–11 GHz.